

- de Hautefeuille, Abbé Jean 126  
*De Humani Corporis Fabrica* 76, 79  
 de Jussieu, Antoine 134, 135, 160  
 de l'Écluse, Charles 88  
 de la Condamine, Charles 139, 143  
 de la Rue, Warren 207  
 de Liuzzi, Mondino 64, 65, 78  
*De Magnete* 94  
 de Maricourt, Pierre 61  
 de Martinville, Édouard-Léon 207  
 de Mondeville, Henri 64, 65  
 de Munck, Jan 141  
 de Nemo, Jordanus 59  
 de Peiresc, Nicolas-Claude Fabri 97  
*De Re Anatomica* 82, 83  
 de Réaumur, René 128, **134**, 138, 373  
*De Revolutionibus Orbium Coelestium* 76  
 de Rivaz, François Isaac 173  
 de Rozier, Pilâtre 156  
 de Saussure, Théodore 172  
 de Thury, César-François Cassini 141  
 de Tournefort, Joseph Pitton 123  
 de Vries, Hugo 236, 237  
   biography 374  
 dead reckoning 130  
 deadbeat escapement 130  
 decay, radioactive 237  
 decimal measurements 163  
 decimal notation 37  
 decimal numbers 184  
 decimal place 29  
 decimal SI prefixes 352  
 decimal system 47, 92  
   in Europe 49  
 decomposition reaction 175  
 Deep Blue chess computer 332  
 Deep Impact 341  
 deep sea drilling 293  
 deep space imaging 349  
 deep space temperature 272  
 Deep/Ultra Deep Field 340  
*Deepsea Challenger* 293  
 degree Celsius, definition 353  
 Deimos 217  
 Deiters, Otto Friedrich Karl 209  
 Delbrück, Max 273  
   biography 369  
 Delisle, Joseph-Nicholas 139  
 dela Porta, Giambattista 83, 86  
 dementia 237  
   Alzheimer's report on 243  
 dementia praecox 228  
 Democritus 24, 99  
 Demosthenes Philalithes 36  
 dendrochronology 336  
 Denis, Jean-Baptiste 112  
 density  
   equations 356  
   units of measurement 353  
 dentistry 118  
   modern 136  
 deoxyribonucleic acid *see* DNA  
 Descartes, René 26, 27, 104, 106  
   biography 369  
   *Principles* 105  
 Desnoyers, Jules 188  
 Desormeaux, Antoine 201  
 deuterium **264**  
 Devil's toenail 169  
 Devonian period 367  
 DeVries, William 314  
 Dewar, James 225, 227  
 Dewar flask 227  
 diabetes 36, 225  
 diagnosis, medical 128  
*Dialogue Concerning Two Chief World Systems* 103  
*Diamond Sutra* 47, 56  
 diamond 166  
 diastase 190  
 diatomic molecule 174  
 Dick, Allen 115  
 Diderot, Denis 143  
 diesel engine 220, 227  
 Diesel, Rudolf 171, 227  
   biography 369  
 dietetics 38  
 Difference Engine 182, 185, 190  
 diffraction 179, 224  
 diffusion 195  
 Digges, Leonard 81, 131  
 digital photography 306, 307  
 digital clock 109  
 digital recording of sound 218  
 digital sound 219  
 digital watch 303  
 dikes 14  
 dilation, time 244  
 dimensions, extra 330  
 Dinoire, Isabele 341  
 dinosaurs 182, 183  
   extinction 311, 324, 325  
   fossils 165  
   naming 196  
   Owen 373  
   soft tissue 341  
 Diocles of Carystus 29  
 diode, Fleming's 239  
 diode valve 243  
 Dione 117  
 Diophantus of Alexandria 38, 46  
   biography 369  
 dioptra 31  
 Dioscorides 33  
 diphtheria 142, 186, 187, 223  
 diploid 226  
 Dirac, Paul 258, **262**, 263, 264, 369  
 discharge tube, first 202  
 disk wheel 20, 21  
 disk, compact 314  
 disk, floppy 302  
 displacement reaction 175  
 dissectible condenser 149  
 dissection 93  
   early **64**, 65  
 distillation **47**, 48  
 diving bell, first 131  
 Diviš, Prokop 144  
 division, mathematical 40  
 division, cell 194, 285  
   *see also* meiosis, mitosis  
 DNA (deoxyribonucleic acid) 194, 279, 282, **284–85**  
   building sequences 319  
   cloning 331  
   Crick 369  
   discovery 211, 369, 374  
   double helix structure 282  
   genetic code 295  
   junk 303  
   noncoding 303  
   nucleotide composition 254  
   recombinant 303  
   replication 288  
   Sanger 373  
   transforming nature of **274**  
   Watson 374  
   X-ray crystallography 270  
 DNA profiling 301, 318, **319**  
 Dobzhansky, Theodosius 270  
 dodo 94  
 Doering, William 274  
 Dokuchaev, Vasily 220  
 Dollond, John 369  
 Dolly 331  
*Dolomedes* 164  
 Domagk, Gerhard 268  
 domains, taxonomy 360  
 domestication  
   animals 13, 14  
   plants 13  
 Dominus of Larissa 39  
 Donald, Ian 288  
 Dong Fang Hong 1 300  
 Doppler, Christian 196, 197  
   biography 369  
 Doppler effect **196**, 225, 369  
 Dornberger, Walter 273  
 double helix (DNA) 284, 285  
 double slit experiment 168  
 Douglass, Andrew 263  
 Dragon, spacecraft 347  
 Drebbel, Cornelius 96, 98, 99  
 Dresden codex 39  
*Drosophila* 318  
   genome sequence 337  
 du Châtelet, Émilie 139  
 du Fay, Charles François de Cisternay 138  
 Dubois, Eugene 227  
 Duchenne, Guillaume 369  
 Duclaux, Emile 221  
 Dudley, Robert 106  
 Dulong, Pierre 180  
 Dulong-Petit law 180, 189, 373  
 Dumbo octopod 293  
 Dunlop, John Boyd 225  
 Durand, Peter 177  
 Dürer, Albrecht 72  
 Durkheim, Emile 229  
 Dutrochet, Henri 192  
 Dutton, Joseph Everett 237  
 Duxbury, Tom 336  
 DVD player 331  
 DVDs 342  
 dwarf planet 168  
 Pluto 342  
 dyeing 29  
 dyes, synthetic, Perkin 373  
 dynamics 28  
 dynamite 210  
   Nobel 372  
 DynaTAC 8000x 315
- E**  
 E = mc<sup>2</sup> 242, 244  
 e-mail 302  
 Eagle Nebula 330  
 Earth  
   age of 130, 156, 209, 249, 286  
   average global temperature 336  
   clues to core 242  
   density 166, 201  
   distance to Sun 113  
   earliest life 348  
   geographic poles **94**  
   geological history 148  
   layers 272  
   magnetic poles **94**  
   magnetic variations 172  
   mass 166  
   measuring 98  
   orbit 100, 101  
   planets 364  
   rotation of 65, 67  
   size of 31, 32, 138  
 Earth Day 300, 301  
 Earth Summit, Rio 328  
 Earth-Sun distance 28  
 earthquakes 36, 148, 221, 268  
   Lisbon 144, 145  
   prediction 374  
   scientific explanation 126  
   waves 242, 243  
 Eastman, George 225  
 Ebbinghaus, Hermann 223  
 Ebers papyrus 22  
 Ebola virus 307  
 ECD 287  
 echo sounding 253  
 echolocation 270  
 eclipse 43  
   first photograph 207  
   solar 61  
 Zhang Heng 374  
 ecological niche 253  
 ecology 232, 374  
 ecosystem 268  
 Eddington, Arthur 242, 245, 254, 255, 256, 257  
   biography 369  
 Edelman, Gerald 291  
 Edison, Thomas Alva 218, 220, **221**, 228, 369  
   light bulb 176  
   phonograph 217  
 Edison effect 221  
*Edmontosaurus* fossils 165  
 EDSAC computer 280  
 education 39  
 Edwards, Robert 308  
 Edwin Smith papyrus 22  
 effort, machine 34, 35  
 egg, mammalian 107, 186  
 Ehman, Jerry 308  
 Ehrlich, Paul 221, 232, 233, 243, **247**, 369  
   syphilis 246  
 Eiffel, Gustav 225  
 Eiffel Tower 225  
 Eijkman, Christiaan 233  
 Einhorn, Alfred 242  
 einkorn 13  
 Einstein, Albert 235, **242**, 257, 369  
   and atomic bomb 271  
   and cosmology 344  
   and relativity 244, 252, 356  
   general relativity 242, **244–45**, **254**  
   gravitational lensing 310  
   miracle year **242**  
   Nobel Prize 255  
   special relativity 242, **244–45**  
 Einstein Observatory 309  
 Einthoven, Willem 238  
 El Niño 297  
   one of worst 314  
 elasticity, Hooke's Law 116  
 electric cell 167  
 electric charge, conduction **136**, 137  
 electric field 234  
   and electromagnetic radiation 234, 235  
 electric light bulb 176, 207, 220  
 Edison 176  
   incandescent 176, 207  
   Swan 374  
 electric locomotive 220, 221  
 electric motor 187  
   principle 182  
 electrical conduction **136**, 137  
 electrical machine, first 127  
 electricity  
   and brain 214  
   chemical 161  
   discovery 161  
   laws 357  
   Ohm 373  
   speed of 200  
   static **128**  
   two kinds 138  
   units of measurements 352, 353, 354  
   Volta 374  
 electrocardiograph 238  
 electrolysis 167, 175, 190, **191**  
   Davy 369  
   of potassium 173  
   of water 173  
 electromagnet 186  
 electromagnetic induction **188**, 189  
 electromagnetic radiation **234–35**  
   Lorentz 372  
 electromagnetic spectrum 234–35  
   early ideas 198  
   Kelvin 374  
 electromagnetic theory, Biot 368  
 electromagnetism 215  
   Maxwell's advances 208  
   Ørsted 181, 373  
   Thomson 374  
 electromagnets **181**  
   Henry's 188  
 electromotive force 190  
 electron capture detector 287  
 electron linear accelerator 278  
 electron microscope 114, 115, 264, 265  
 electronic calculator 185, 302  
 electronic mail 302  
 electronics, start of 239  
 electrons 250, 251, 266, 267  
   and cosmology 345  
   and wave-particle duality 257  
   charge on 243  
   coining of term 227  
   discovery of 232  
   Standard Model 304  
 electrophorus 153  
 electroscope 95  
 electrostatic force 266  
 electrostatic generator 149  
 electroweak force 296  
 electroweak theory  
   Salam 373  
   Weinberg 374  
 elements 358–59  
   and atoms 250  
   and compounds 174  
   atomic theory 178  
   Berzelius 368  
   classification 48  
   Dalton's table 172  
   essential information 359  
   Moseley 372  
   periodic table 210, 211, 358  
   tables of 160, 358, 359  
   theory of four 25  
   transuranium 268  
 elevator (lift), first 203  
 Elhuyar, José and Fausto 156  
 elliptical orbits 100, 101  
 embryology 95, 102, 187  
 embryonic stem cells, human 333  
 embryos, genetic testing of 323  
 emmer 13  
 Empedocles 25  
 enameling 16, 23  
 Enceladus 160  
*Endeavour* 151, 292  
 endoscope 201  
   brass 91  
 endoscopy 182  
   wireless 330  
 endosymbiotic theory 295, 300  
 energetic reaction 174  
 energy  
   and chemical reaction 256  
   and mass 242, 244  
   conservation of **199**  
   dark 333  
   Joule 352, 371  
   units of measurement 352, 353, 354, 371  
 energy budget, Earth's 327  
 energy packages, quanta 236  
 Enewetak Atoll 282  
 engineering, Brunel 369  
 engines **170–71**  
   atmospheric 122  
   Diesel 369